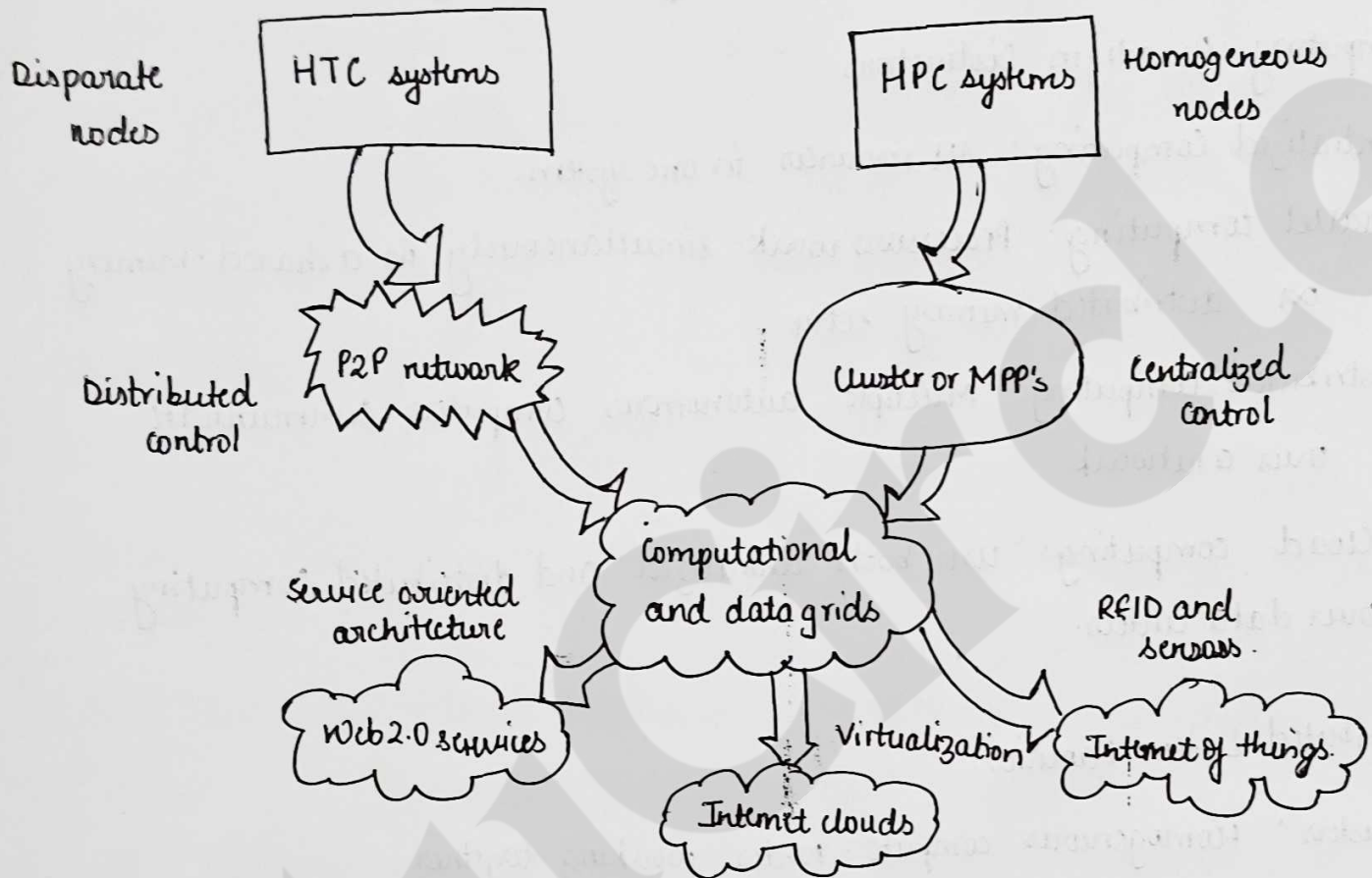


Cloud Computing

Module 1

① Illustrate the scalable computing over the internet.

→



1) High Performance Computing (HPC):

- * Focused on raw speed, measured in floating point operations per seconds.
- * Used mainly in scientific, engineering and industrial applications.
- * Limited to a small number of specialized users.

2) High Throughput Computing (HTC)

- * Shift from HPC to HTC for market oriented applications.
- * Used in Internet searchers, web services and enterprise computing.
- * Emphasis on cost reduction, energy efficiency, security and reliability.

3) Emerging Computing Paradigms

- * Service oriented Architecture : Enables Web 2.0 services.
- * Virtualization: Key technology for cloud computing
- * Internet of things: Enabled by RFID, GPS and sensor technologies
- * Cloud Computing: Evolution of computing as a utility.

4) Computing Paradigm Distinctions:

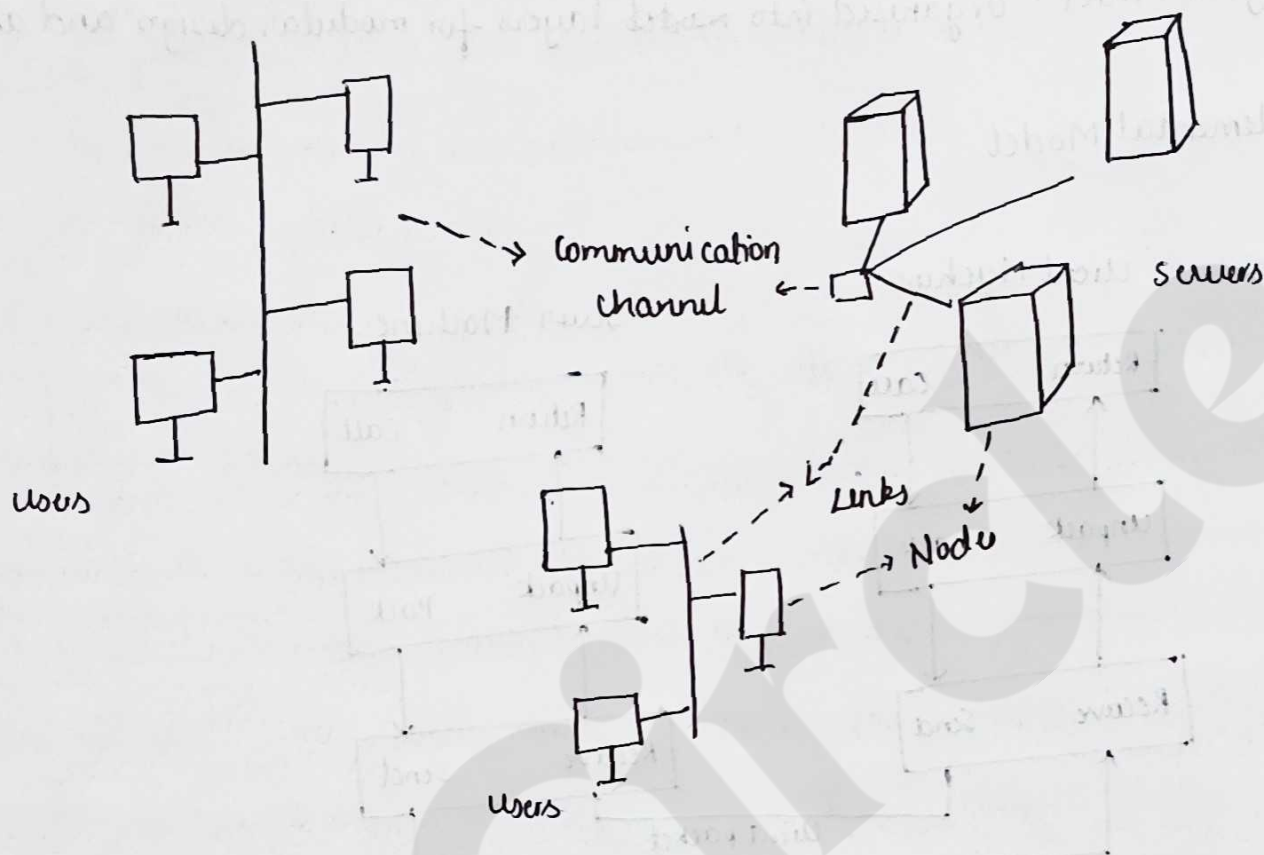
- * Centralized Computing: All resources in one system.
- * Parallel computing: Processors work simultaneously in a shared-memory or distributed memory setup
- * Distributed Computing: Multiple autonomous computers communicate over a network
- * Cloud computing: uses both centralized and distributed computing over data centers.

5) Distributed System Families:

- * Clusters: Homogeneous compute nodes working together.
- * Grids: Wide area distributed computing infrastructures.
- * P2P networks: Client machines globally distributed for file sharing and content delivery.
- * Cloud Computing: utilizes clusters, grids and P2P technologies.

② Discuss in detail about distributed system models.

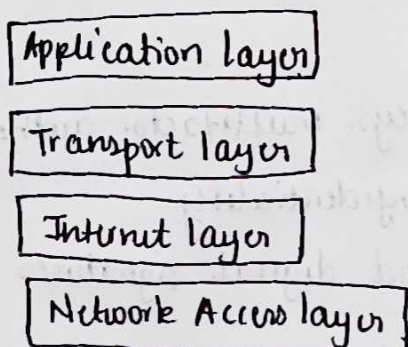
→ 1) Physical Model:



Physical Model Represents the hardware layout of the system

- a) Nodes: Devices that process and communicate
- b) Links: Communication channel like point to point communication
- c) Middleware: Software that enables communication, fault tolerance, synchronization
- d) Network topology: Structure of node connections.
- e) Protocols: TCP, UDP, MQTT

2) Architectural Model:



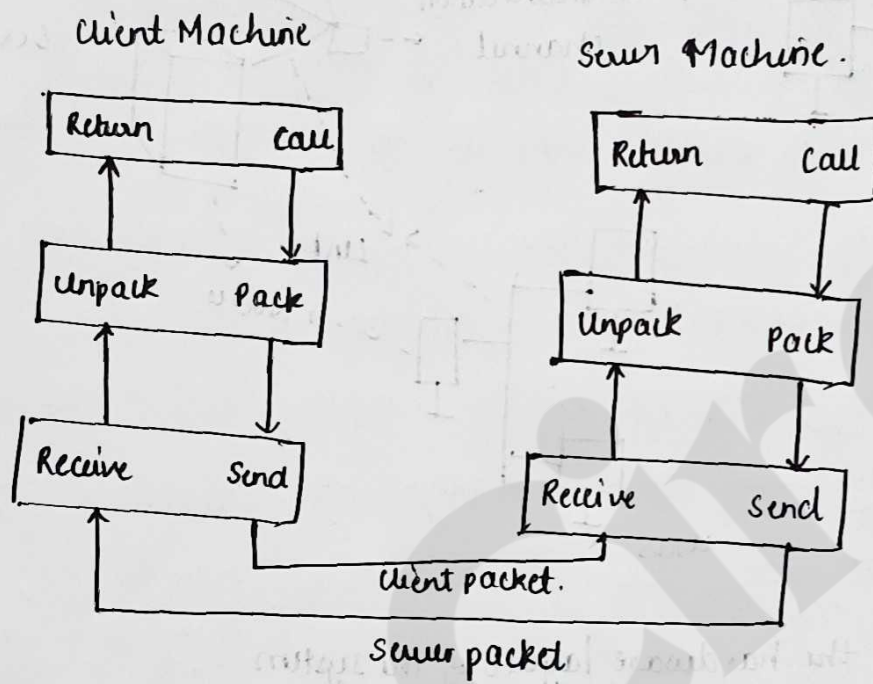
Architectural Model defines the system organization and interaction patterns

a) Client Server Model: Centralized server responds to client request.

b) Peer to Peer: All nodes are equal and share services.

c) Layered Model: Organised into ~~model~~ layers for modular design and abstraction

3) Fundamental Model:



Fundamental Model covers key concepts and formal behaviour.

a) Interaction Model:

Message passing: Synchronous / asynchronous communication.

b) Failure Model:

Types: Crash, omission, timing, Byzantine failures

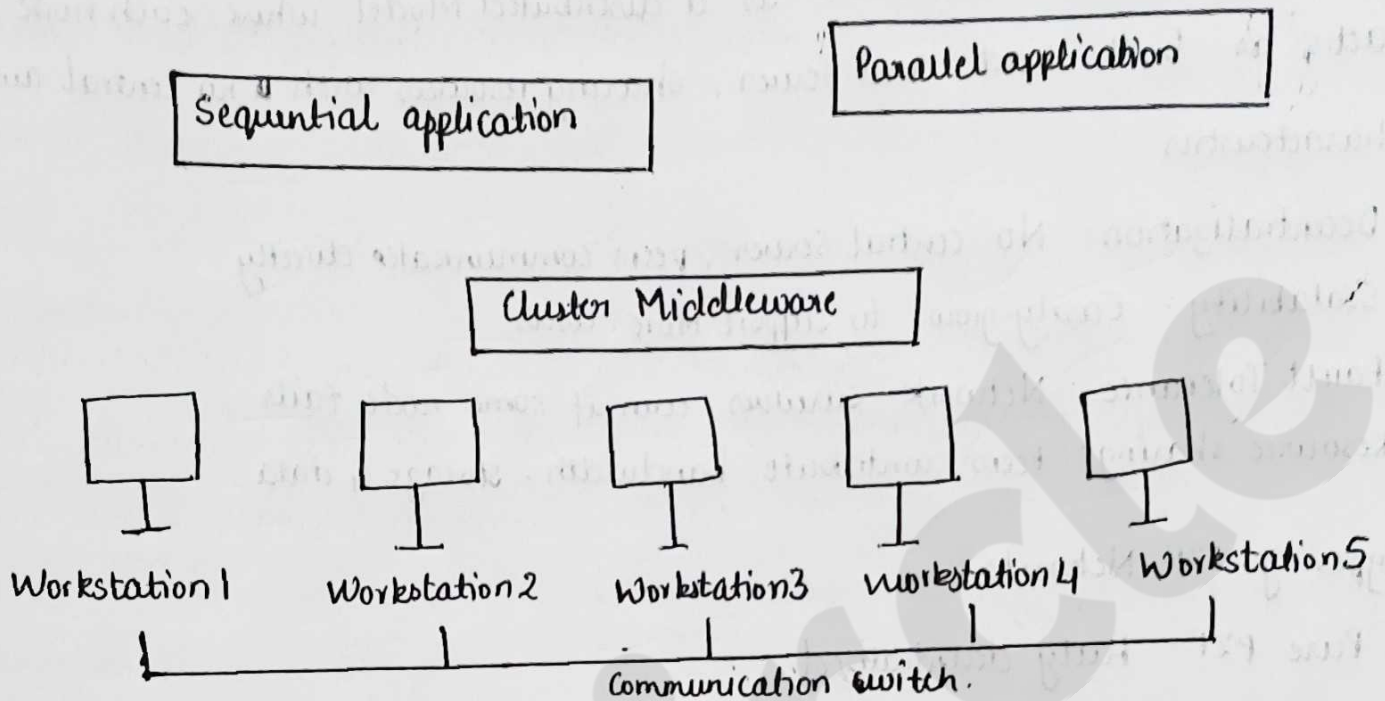
c) Security Model

1) Authentication: Passwords, keys, multifactor authentication

2) Encryption: Protects data confidentiality

3) Data Integrity: Hashing and digital signatures to prevent tampering

③ Explain the Basic cluster architecture with a neat diagram.



- 1) Cluster Computing is a technique where multiple interconnected computers work together as a single system to execute tasks, process data and run applications.
- 2) Node : Each node has its own processor, memory and OS.
- 3) Head Node: Manages the cluster operations, distributes tasks to other nodes
- 4) Slave Node : Execute assigned task and report back to the head node
- 5) Cluster Middleware: Software that handles job scheduling, communication and resource management.
- 6) Interconnect Network: Ensures high speed data transfer between nodes.
- 7) Storage System: shared or distributed storage system, all nodes may access the same data.

④ Write short note on Peer to Peer network families.

→

1) P2P ~~architecture~~ architecture is a distributed Model where each node acts as both client and server, sharing resources with no central authority.

Characteristics

- 2) Decentralization: No central server, peers communicate directly
- 3) Scalability: Easily grows to support more users
- 4) Fault Tolerance: Network survives even if some node fails
- 5) Resource sharing: Peers contribute bandwidth, storage & data

Types of P2P Networks

- 6) Pure P2P: Fully decentralized
- 7) Hybrid P2P: Uses central servers or superpeers
- 8) Overlay P2P: Virtual network over physical internet
- 9) Structured P2P: Organized topology with routing rules
- 10) Unstructured P2P: Random topology, no fixed structure

Components

- 11) Peer Nodes: Active devices in Network
- 12) Overlay Network: Virtual layer connecting peers
- 13) Indexing Mechanisms: Help locate shared resources
- 14) Bootstrapping Mechanisms: Enable new peers to join the network

⑤ Discuss system attacks and threats to cyberspace resulting in 4 types of losses

→ Common System Attacks

- 1) Malware attacks: Includes viruses, worms, spyware and Trojans
- 2) Phishing: Deceptive messages to trick users into giving up sensitive info like passwords
- 3) Denial of Service: Overload networks or servers, making them unavailable to users.
- 4) Man in Middle attack: Attackers intercept communication between two ~~parties~~ parties to steal or alter data
- 5) SQL Injection: Injects malicious SQL queries into input fields to access or manipulate databases.

Four types of loss due to cyber attacks.

- 1) Financial Loss: Cyber attacks like ransomware or online fraud can lead to direct theft of money or demand for large payments.
- 2) Data loss: Attacks such as malware, hacking or unauthorized access can result in loss or theft of sensitive data
- 3) Reputational Loss: A successful cyber attack damages an organization's public image and brand value
- 4) Operational Loss: Cyber threats like Denial of Service can bring down servers, disrupting business operations.

⑥ Compare ~~cluster~~ architecture cluster computing and peer to peer system in context of cloud architecture

→

Cluster Computing

- 1) Centralized architecture
- 2) Controlled by a central manager
- 3) Resources are Pre allocated and fixed
- 4) Moderate Fault Tolerance
- 5) Limited scalability
- 6) Point to Point Communication model
- 7) Used for data centers and high performance computing
- 8) Ex : Apache Hadoop clusters

Peer to Peer system

Decentralized architecture

No central controller

Resources are dynamic and voluntary

High fault Tolerance

Highly scalability

Peer to Peer

Use in distributed storage, file sharing

Ex: BitTorrent.

⑧ Explain briefly about Technologies for network based system.

→ 1) Broadband networks and Internet architecture:

* High speed internet access is crucial for cloud computing. Broadband networks supports data transmission over the internet with low latency and high throughput.

2) Data center technology:

Cloud computing is powered by large scale data centers which house numerous servers and storage system.

These data centers use virtualization and load balancing to manage resource efficiently.

3) Virtualization Technology:

Virtualization allows multiple VMs to run on a single physical machine. It abstracts hardware resource and provides scalability and isolation for cloud applications.

4) Web 2.0 and Service oriented Architecture (SOA):

Web 2.0 technologies allows interactive and collaborative web applications.

SOA allows different services to communicate over a network using standard protocols, making integration across system seamless.

5) Webservices and Internet Protocols:

Cloud platforms use webservices for communication between distributed system.

Standard protocols such as HTTP, TCP/IP, and DNS support the delivery of cloud services over the internet.

6) Distributed Computing Models:

Grid computing, utility computing and distributed system are precursors to modern cloud computing.

7) ~~Auto~~ Autonomic computing: System manage themselves with minimal human intervention. Features include self configuration, self-healing, self optimization.

8) Client-Server computing.

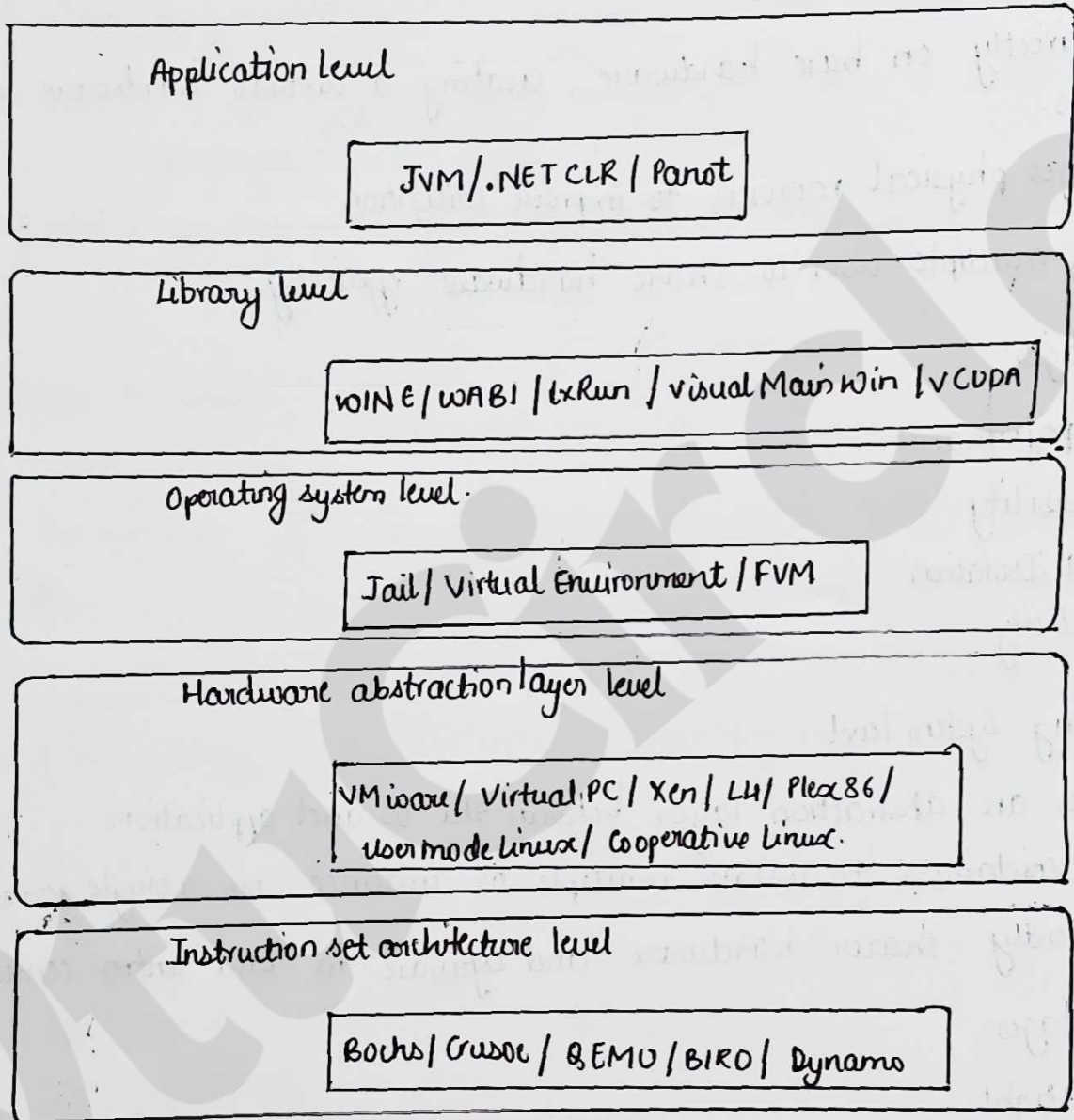
The traditional model is extended in the cloud with thin clients and web based interfaces. interacting with powerful cloud servers.

Cloud Computing

Module 2

① Explain the different level of virtualization and their advantages.

→



i) Instruction set architecture level:

- * ISA virtualization emulates one instruction set architecture on another.
- * Enables running legacy binary code on different hardware.
- * Basic method: Code interpretation - translating each instruction individually
- * Optimized method: Dynamic binary translation - converts block of instruction for better performance

Advantages

- 1) Legacy code support
- 2) Cross architecture compatibility
- 3) Facilitates Hardware upgrades.

2) Hardware Abstraction level:

- * Runs directly on bare hardware, creating a virtual hardware environment for VMs.
- * Manages physical resources to improve utilization.
- * Allows multiple user to share hardware efficiently.

Advantages:

- 1) High performance
- 2) Scalability
- 3) Fault Isolation.
- 4) Versatility.

3) Operating System level:

- * Creates an abstraction layer between the OS and application
- * Uses containers to isolate multiple OS instance on a single server
- * Efficiently shares hardware and software in data centers.

Advantages:

- 1) Light weight
- 2) High performance
- 3) Scalability
- 4) Isolation
- 5) Ease of deployment

4) Library support level:

- * Virtualizes API's instead of system calls for better flexibility.
- * Controls communication between application and system using API hooks.

Advantages:

- 1) API Hooks
- 2) Controlled Communication
- 3) Application specific Virtualization.

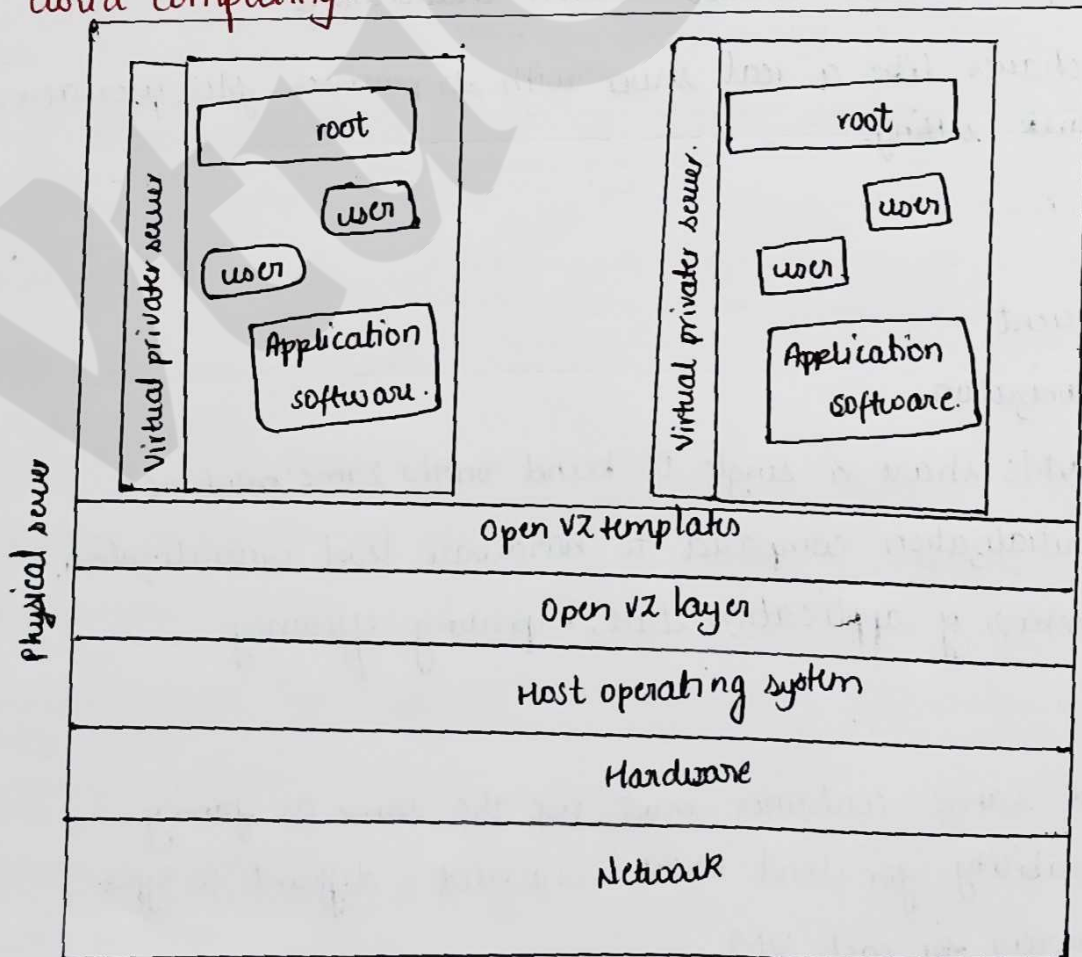
5) user application level Virtualization.

- * virtualizes application as VMs also ~~called~~ called process level virtualization
- * It uses highlevel language VMs, such as JVM and .NET CLR
- * Creates an abstraction layer that allows programs to run on system with the VM.

Advantages

- 1) Cross Platform Compatibility
- 2) Improved Security
- 3) Simplified Deployment
- 4) Portability

② Describe OS level virtualization what are advantages and limitations in cloud computing.



- 1) Cloud computing leverages VM technology to reduce hardware and staffing costs by outsourcing to third parties.
 - 2) Dynamic resource allocation: Workload vary, needing different number of VM at different times.
 - 3) slow VM instantiation: New VMs start from fresh boots or templates without awareness of current application state.
- Issues with Hardware level virtualization.
- 4) slow VM initialization: Each VM starts from scratch, which is inefficient for large scale cloud computing.
 - 5) storage concerns: Many VM images contain redundant data.
 - 6) Performance overhead: Full virtualization is slower, has lower density, and may require ~~many~~ modifying the guest OS.

- OS level virtualization as solution:

- 7) Inserts a virtualization layer inside the OS to partition resources.
- 8) Allows multiple isolated Virtual Execution Environment.
- 9) Each VE behaves like a real server with its processes, files, user accounts and network settings.

Advantages:

- 1) Fast and efficient
- 2) State synchronization
- 3) All OS-level VMs share a single OS kernel on the same machine.
- 4) Faster VM initialization compared to hardware level virtualization.
- 5) Better awareness of application state, improving efficiency.

Limitations

- 1) All VMs on the same container must use the same OS family
- 2) This limits flexibility for cloud users who need different OS types.
- 3) Duplicate resources for each VM.

③ How do virtual cluster contribute to cloud resource management.

→ A virtual cluster consists of multiple virtual machines that are grouped logically and work together to perform computational tasks.

Contribution of cloud Resource Management:

- 1) Dynamic Resource Allocation: Resources like CPU, memory, and storage can be assigned dynamically based on current demand.
- 2) Load Balancing: Tasks can be evenly distributed across VMs to avoid overloading any single node.
- 3) High availability: If one VM fails, the workload can be migrated to another VM in the cluster.
- 4) Energy efficiency: Idle physical machines can be powered down or consolidated.
- 5) Scalability: VMs can be added or removed from the cluster depending on demand, enabling elastic scaling.

④ Evaluate security risks associated with virtualization and propose mitigation strategies.

→ Security Risk:

- 1) VM Escape: A malicious user or process break out of the VM to access the host system.
- 2) Hypervisor attacks: Exploiting vulnerabilities in the hypervisor to gain control over all hosted VMs.
- 3) Insecure VM Images: Use of unverified or tampered images could lead to security breaches.

4) Inter VM attacks: A compromised VM can attack other VMs on the same host if proper isolation isn't maintained.

5) Data leakage: Shared resources may lead to unintended data access.

Mitigation Strategies:

1) Patch Management: Regularly update hypervisors and guest OS to fix vulnerabilities.

2) Access Control: Use strict role-based access control to limit admin privileges.

3) Secure configuration: Harden VM templates and remove unnecessary services.

4) Network Isolation: Use virtual networks and firewalls to isolate traffic between VMs.

5) Monitoring and logging: Implement security monitoring, intrusion detection system and maintain logs for audits.

⑤ What is hardware-assisted virtualization? Discuss how Intel VT-x and AMD-V enhance virtualization performance.

→ Hardware assisted virtualization is a method where CPU features help to improve the performance and efficiency of virtualization by allowing the hypervisor to run directly on the hardware.

Intel VT-x and AMD-V are virtualization extensions provided by Intel and AMD respectively. They allow VMs to run with fewer overheads and better isolation.

- 1) Privilege Levels: They introduce a new CPU mode that separates VM execution from hypervisor.
- 2) Reduced Overhead: Hardware traps are used instead of binary translation for sensitive operations, improving speed.
- 3) Nested Virtualization: Support running hypervisors inside VMs useful for testing and lab environments.
- 4) Better Security: Hardware enforced isolation prevents VMs from accessing host resources directly.

⑥ Discuss role of CPU, memory, I/O virtualization in optimizing cloud performance.

→ CPU virtualization:

- 1) Enables multiple VMs to share same physical processor
- 2) ~~Help~~ Helps in fair scheduling and dynamic CPU allocation
- 3) Improves multitasking and workload balancing.

Memory virtualization:

- 1) Allows dynamic memory allocation and overcommitment.
- 2) Uses techniques like ballooning, memory swapping and shared memory.
- 3) Enhances performance by making better use of available physical memory.

I/O virtualization:

- 1) Abstracts input/output devices like disks and network interfaces.
- 2) Reduces I/O bottlenecks using technologies like SR-IOV (Single Root-I/O Virtualization)
- 3) Enables efficient sharing of physical devices with low latency.

⑦ Compare Type 1 and Type 2 hypervisors with real world problems. ⑧

→ Type 1 hypervisor	Type 2 hypervisor
1) Bare metal hypervisor	Hosted hypervisor
2) Runs directly on physical hardware	Runs on top of a host OS
3) High performance, due to direct hardware access	Low performance, as it depends on host OS.
4) More secure and isolated	Less secure and depend on host OS
5) High resource efficiency	Low resource efficiency
6) Ex: VMware ESXi, Microsoft Hyper-V, Xen	Oracle, VirtualBox, VMware Workstation

⑧ Explain how Migration of Memory, Files, and network resources happen in cloud computing.

-
- 1) Memory Migration:
 - a) VM memory migration involves memory states efficiently
 - b) The Internet Suspend-Resume technique uses temporal locality to transfer only changed data
 - c) A tree based file structure minimizes data transfer by copying only modified parts.
 - 2) File System Migration:
 - a) VM migration requires a consistent, location independent file system accessible on all hosts.

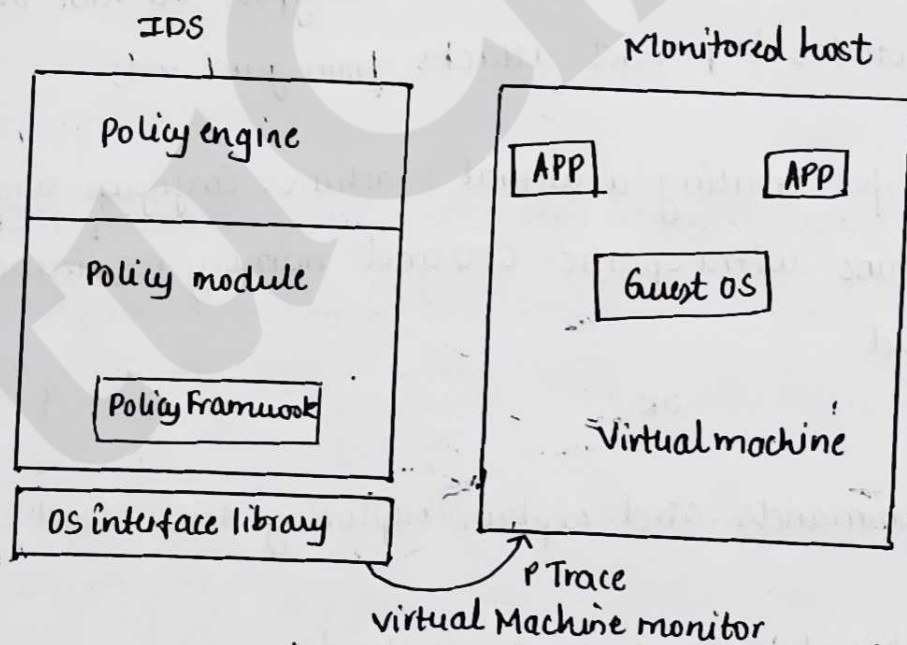
- b) Two approaches:
- Virtual disk transport
 - Global distributed file system.
- c) Smart Copying uses spatial locality, transferring only differences between the old and new locations.

3) Network Migration:

- Migration VMs must maintain network connection without disruption
- Each VM has a virtual IP and MAC address, separate from the host machine
- Switched network detect new VM locations and reroute traffic automatically

9) Explain VM based intrusion detection system.

→



- 1) Intrusion detection system help identify unauthorized access. IDS can be
- Host based IDS:** Runs on a monitored system but is vulnerable to attacks.
 - Network based IDS:** Monitors network traffic but struggles to detect fake actions.

The Virtual Machine Monitor can audit access request, combining the strengths of a HIDS and NIDS. There are two methods of implementation

- 1) IDS as an independent process inside a high privileged VM on the VMM
- 2) IDS integrated into the VMM with direct hardware access.

Granfinkel and Rosenblum proposed a VMM based IDS that monitors guest VMs using a policy framework and trace based security enforcement. However, logs used for analysis can be compromised if operating system is attacked.

Honey pots and Honey nets:

Besides, IDS, honeypots and honey nets are used to detect attack by tricking attackers into interacting with fake system. Honey pots can be physical or virtual, and in virtual honeypots the host OS and VMM must be protected to prevent attacks from guest VMs.

- ⑩ write steps for creating a virtual Machine: Configure and deploy a virtual machine with specific CPU and memory requirements in google ~~cloud~~ cloud.

OR

write 5 commands and explain exploring AWS, cloud shell.

→ creating Virtual Machine in google cloud:

- 1) Go to google cloud console

* URL: <https://console.cloud.google.com>

* Log in to your google account

2) Select or Create project

- * Either select an existing project or create a new one

3) Enable the Compute engine API:

- * Navigate to Compute engine > VM instances

- * If prompted, enable the compute engine API

4) Create a new VM instance

- * Click create Instance

5) Configure your VM:

- * Name: Enter a name for the VM

- * Region and Zone: Choose geographic region & zone

- * Machine Type: Click customize to set CPU, memory

- * Boot disk: Choose the OS

- * Optional: Configure Firewall setting to allow HTTP/HTTPS

6) Create VM

- * Click create button

- * Your virtual machine will be deployed, and ~~you~~ you'll be able to SSH into it once it's ready.

5 commands explaining ~~the~~ aws cloud shell

Command 1: `aws configure`: Configure your AWS CLI with access credentials, you will enter your access ID, secret Access key, default region and output format.

Command 2: ~~aws~~ `aws s3 ls`: List all your S3 buckets, This command helps to verify that you have access to your AWS account's S3 resources

Command 3: `aws ec2 describe instances` : List all EC2 instances, it provides detailed information such as instance ID, status, and type

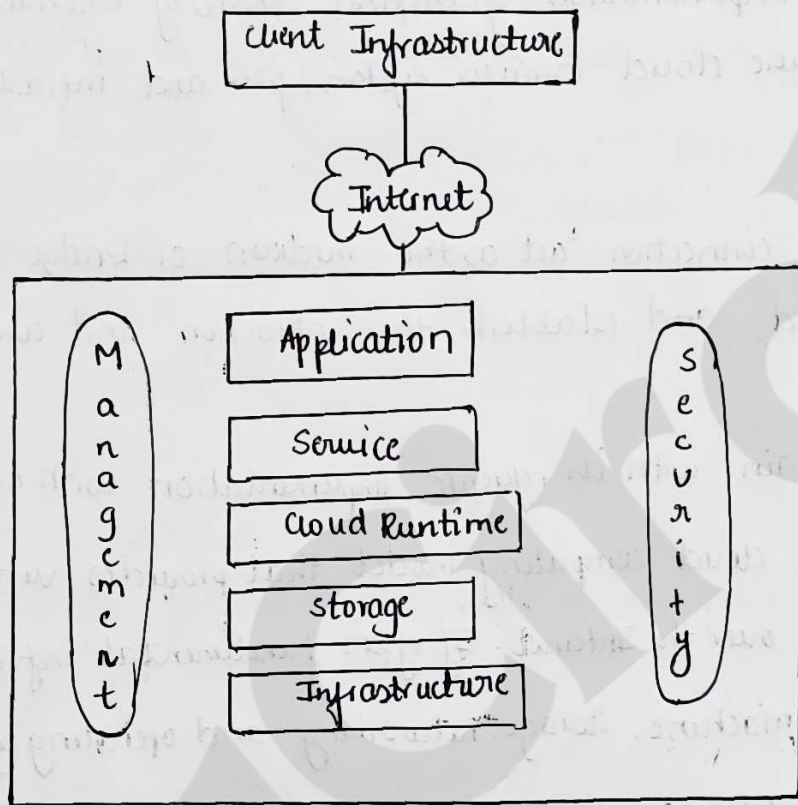
Command 4: `aws lambda list-functions` : List all lambda functions in your account. useful to check serverless functions without going to the aws console

Command 5: `aws help` : Get help on aws cli usage. display documentation on all available aws CLI commands.

Module 3

① Briefly explain cloud computing architecture with neat diagram.

→



- 1) Client Infrastructure: It contains the application and users interface which are required to access the cloud platform.
- 2) Application: Refers to a software or platform to which client accesses.
- 3) Service: Refers to the major three types of cloud-based services like SaaS, PaaS, IaaS. Also manages which type of service the user accesses.
- 4) Runtime cloud: Provides execution and runtime platform/environment to the virtual machine.
- 5) Storage: Provides flexible and scalable storage services and management of stored data.

- 6) Infrastructure: Refers to hardware and software components of cloud like it includes servers, storage, network devices, virtualization software.
- 7) Management: Management of backend components like application, service, runtime cloud, storage, infrastructure and other security mechanism.
- 8) Security: Refers to implementation of different security mechanisms in the backend for secure cloud resources, system, files and infrastructure to end user.
- 9) Internet: Internet connection act as the medium or bridge between frontend & backend and establish the interaction and communication.

② Define IAAS? Explain with its reference implementation with neat diagram.

→ IAAS: IAAS is a cloud computing model that provides virtualized computing resources over the Internet. It offers fundamental infrastructure such as virtual machine, storage, networking and operating systems on a pay-as-you-go basis.

Reference implementation

Amazon Web Services: Hosting an web application using amazon EC2

- 1) User logs into AWS management console
- 2) Launches and EC2 instance
- 3) Choose an OS (Windows/Linux)
- 4) Configure storage, security and network settings
- 5) Deploy web application on the instances.
- 6) Manages and monitors using AWS cloud watch.

③ Discuss IaaS, PaaS, and SaaS cloud service models at different service levels

→ IaaS: Provides essential computing services and resources as virtualized server over internet.

Key characteristics:

- 1) Pay-as-you-go pricing model
- 2) Highly scalable infrastructure
- 3) Automated resource provisioning
- 4) AWS EC2: Provides virtual machine with flexible configuration
- 5) Google Compute Engine: Scalable VM instances for cloud workloads
- 6) Microsoft Azure Virtual Machine: Supports both windows and linux operating system.

PaaS: Provides a cloud based environment for application development and deployment. Includes integrated tools such as databases, runtime environment and development frameworks.

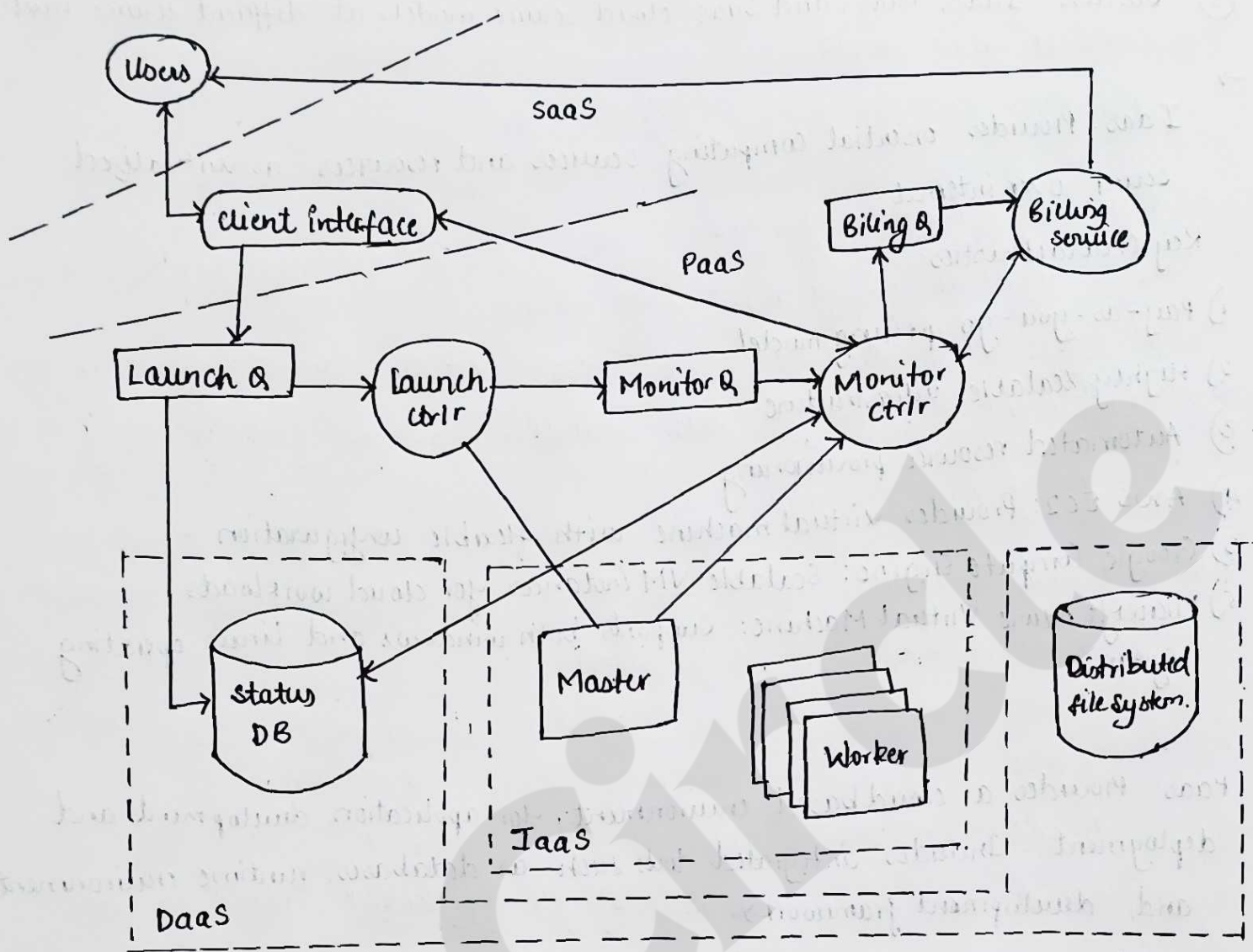
Key characteristics:

- 1) Reduces the complexity of software development
- 2) Allows developers to focus on writing code instead of managing infrastructure
- 3) Google App engine: Scalable application hosting
- 4) AWS elastic Beanstalk: Automated deployment and scaling of web applications
- 5) Microsoft Azure App Services: PaaS offering for .NET, Java, python & more

SaaS: Delivers software applications over the internet without requiring installation on local machine.

Key characteristics

- 1) Users access software via web browsers on a subscription basis
- 2) Google workspace: Cloud based productivity tools
- 3) Microsoft 365: office application and collaboration tools
- 4) Salesforce: Customer relationship management software.



④ Explain private, public and hybrid cloud deployment models. OR Explain different types of cloud.

a) Public cloud:

- 1) Hosted and maintained by third party cloud service provider
- 2) Resources such as computing power, storage and networking are shared among multiple customers
- 3) Offers cost effectiveness, scalability and flexibility.
- 4) Example: Amazon webservices, Microsoft Azure, Google cloud platform.

b) Private cloud

- 1) Dedicated cloud infrastructure for a single organization, either hosted on-premises or by third party provider.
- 2) Provides enhanced security, compliance and performance optimization

- 3) Ideal for industries with strict regulatory requirements such as healthcare and finance.
- 4) High operational costs due to maintenance and hardware requirements.
- 5) Example: VMware vSphere, OpenStack based private clouds.

9) Hybrid cloud

- 1) A combination of public and private cloud infrastructure to optimize performance and cost efficiency.
- 2) Workloads can be dynamically shifted between private and public clouds based on business ideas.
- 3) Benefits include greater flexibility, scalability and disaster recovery options.
- 4) Example: AWS outposts, Microsoft Azure stack.

5) Write a short note on global exchange of cloud resources.

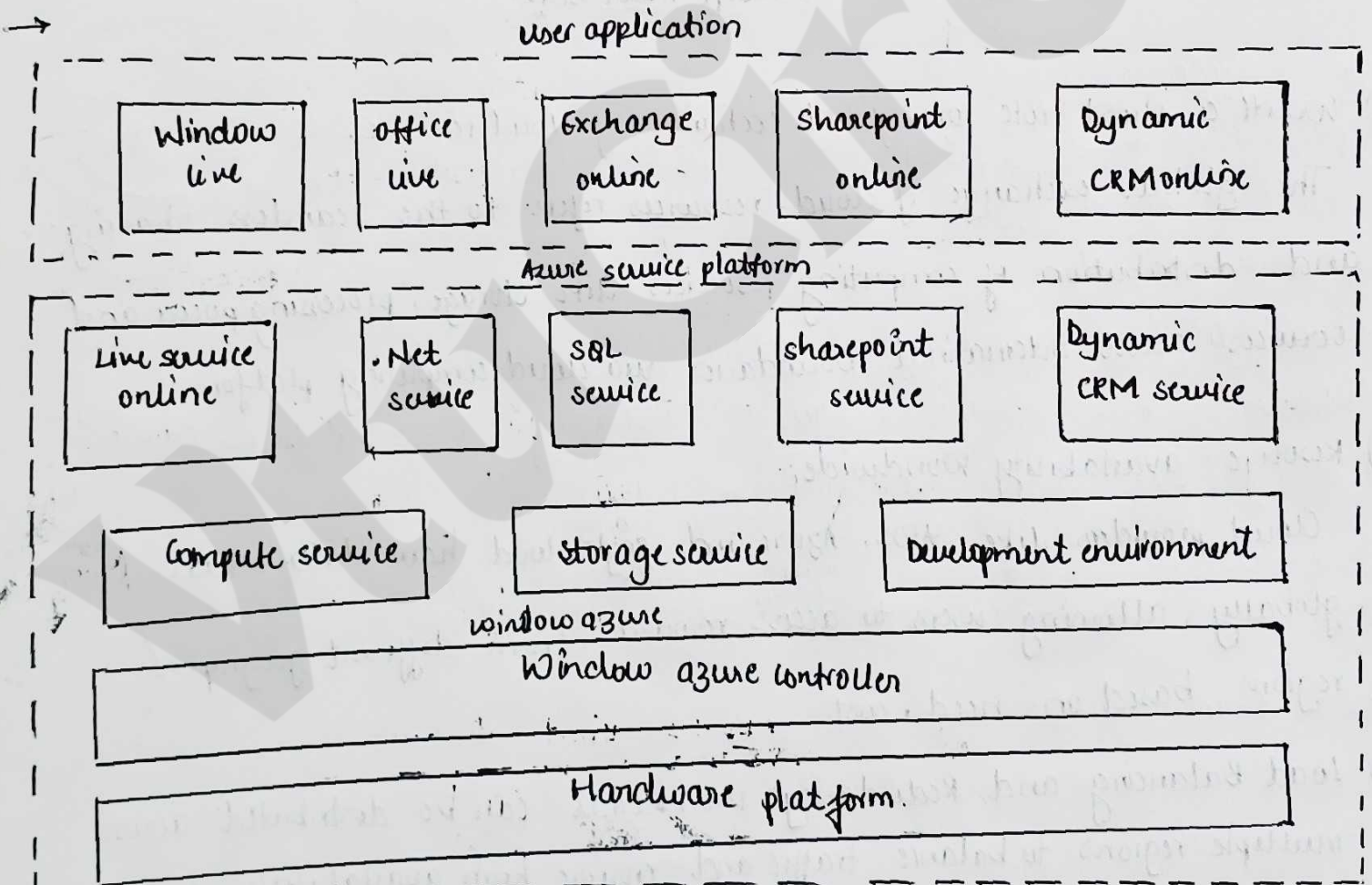
→ The global exchange of cloud resources refers to the seamless sharing and distribution of computing resources like storage, processing power and services across international boundaries via cloud computing platforms.

1) Resource availability worldwide:

Cloud providers like AWS, Azure and Google Cloud have data centers globally, allowing users to access resources from different geographic regions based on need, cost.

- 2) Load Balancing and Redundancy: Workloads can be distributed across multiple regions to balance traffic and ensure high availability, fault tolerance and disaster recovery.
- 3) Latency Optimization: Delivering content and services from data centers closest to the user help reduce latency and improve performance.

- 4) Regulatory Compliance: While exchanging resources globally, cloud providers ensure compliance with data sovereignty laws and privacy regulation like GDPR.
- 5) Cost efficiency: Users can choose regions with cheaper computing costs or off-peak pricing to optimize their cloud spending.
- 6) Collaboration and Scalability: Enables multinational teams to collaborate in real time using shared cloud environments and scale resources as per demand.
- ⑥ Discuss a set of cloud services provided by Microsoft Azure.

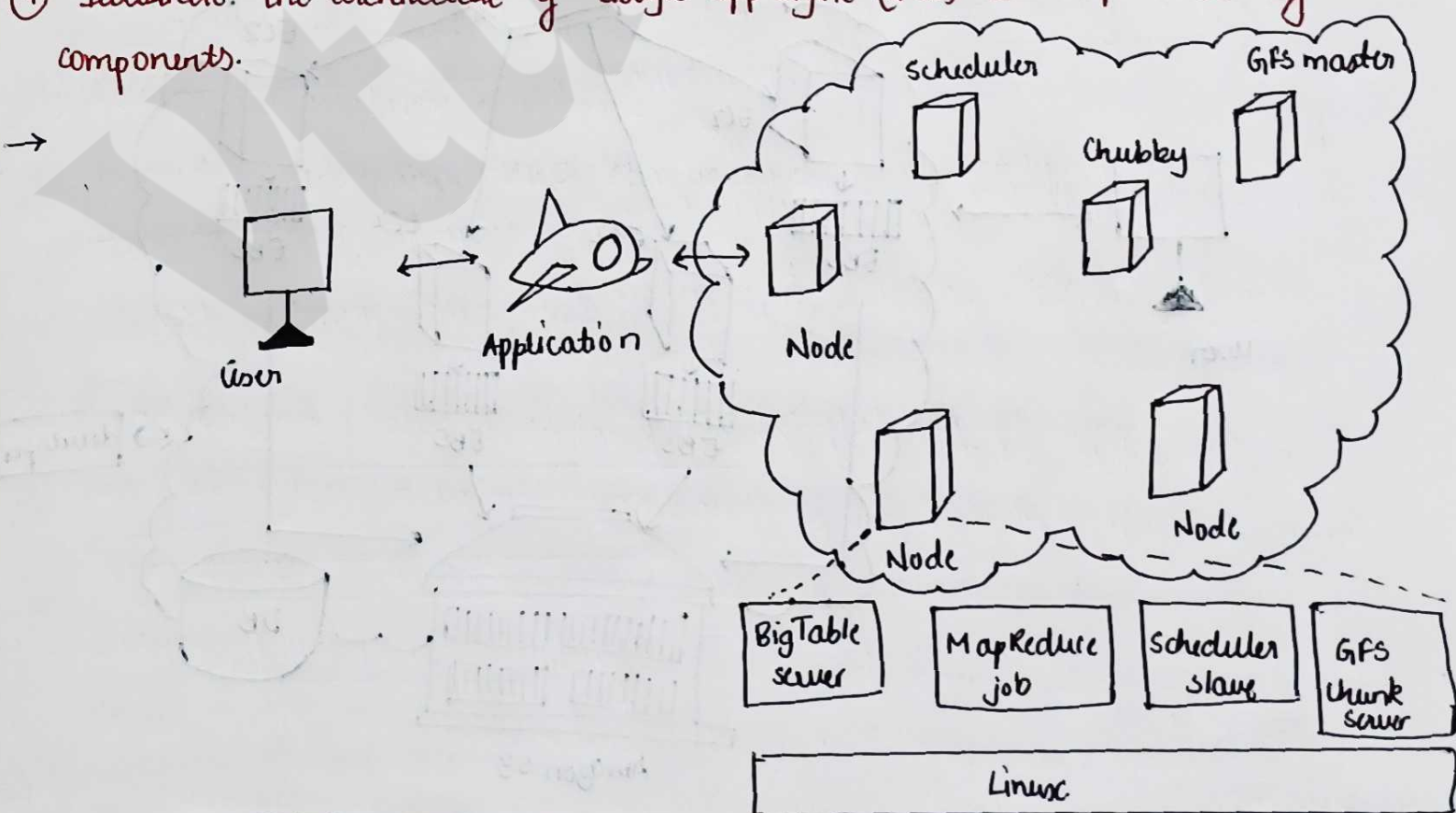


⑦ **Windows Azure:** Provides a cloud platform based on Windows OS, manages VMs, storage, and network resources in data centers.

2) Core Azure Services:

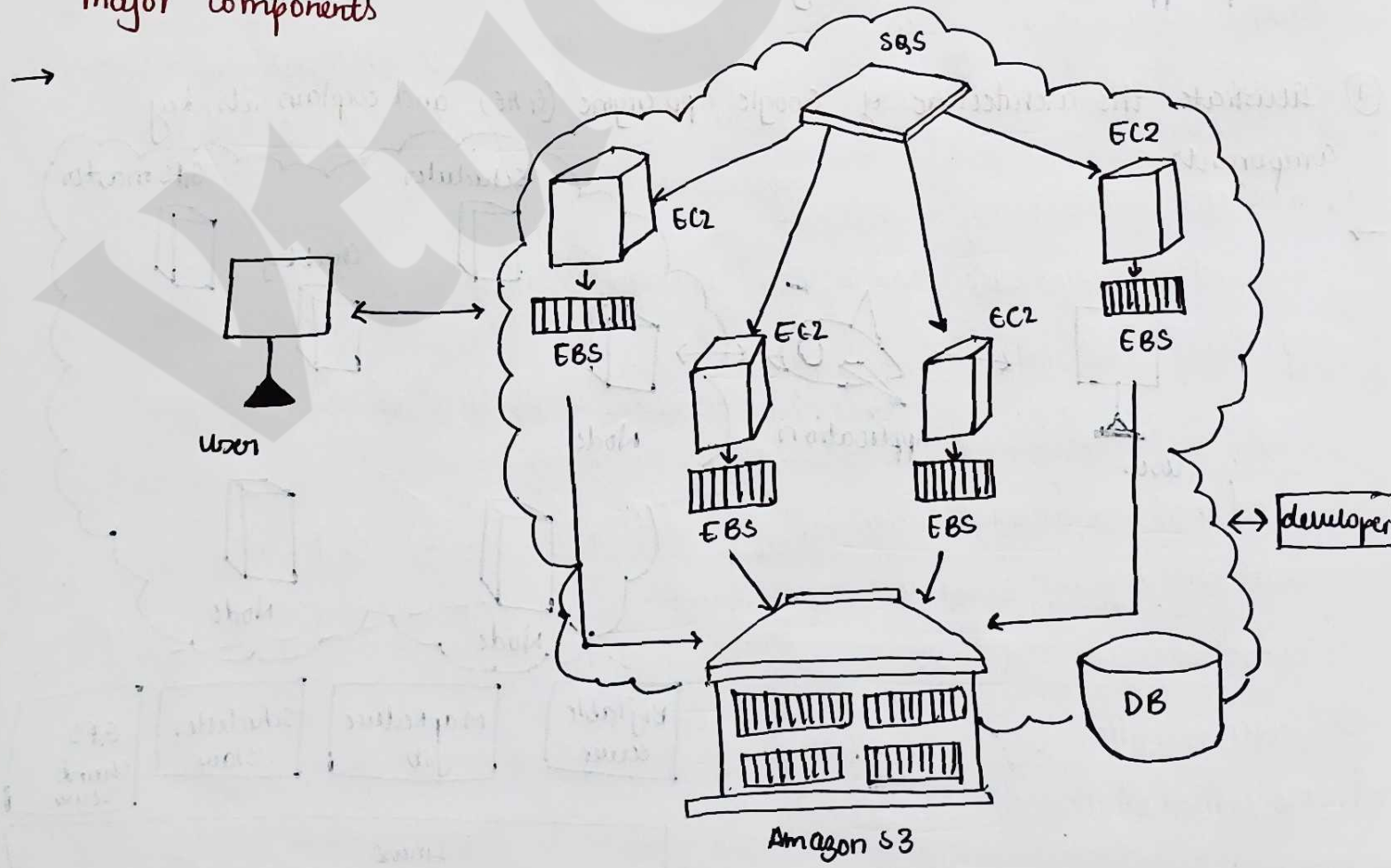
- * **Live Service**: Support Microsoft live application and multimachine data access
 - * **.Net service**: Enables local application development and cloud execution
 - * **SQL Azure**: Provides cloud based relational database service with SQL server
 - * **Sharepoint Service**: Help in scalable business application development
 - * **Dynamic CRM service**: Support business management applications.
- 3) **Integration with Microsoft products**: Azure services work seamlessly with windows live, office live, exchange online, share point online, dynamic CRM online.
- 4) **Communication Protocols**: Uses SOAP and REST for webbased communication and third party cloud integration.
- 5) **Azure development kit**: Allow developers to run a local version of azure, develop applications and debug them on windows hosts.

7) Illustrate the architecture of Google app engine (GAE) and explain its key components.



- 1) Google app engine is a platform as a service that allows developers to build and deploy application on google cloud infrastructure.
- 2) The frontend ~~server~~ serves as an application framework, similar to ASP, J2EE, and JSP supporting python and java for application development
- 3) The GAE infrastructure supports multiple cluster of servers running GFS, MapReduce, BigTable, and chubby to provide efficient resource management.
- 4) Data store: Provides object oriented, distributed, structured data storage based on Bigtable
- 5) Application Runtime environment: Support scalable web programming & execution
- 6) Software development kit: facilitates local application development and testing
- 7) Administration Console: Simplifies the management of application development cycles
- 8) GAE web service Infrastructure: Provides special interfaces for efficient storage and network resource management.

8) with a clear diagram, explain the architecture of AWS and describe its major components



- 1) Amazon Web Service is a leading provider of public cloud services using the Infrastructure as a Service model.
- 2) EC2 (Elastic Compute cloud): Provides virtualized computing platforms for running cloud applications.
- 3) S3 (Simple Storage Service): offers object oriented storage for data management
- 4) EBS (Elastic Block Service): Supports block storage for traditional application
- 5) SQS (Simple Queue Service): Ensures reliable message processing between processes
- 6) ELB (Elastic Load Balancing): Distribute incoming traffic across multiple EC2 instances.
- 7) Cloud Watch: Monitors AWS resources.
- 8) RDS (Relational database service): Provides fully managed relational database service
- 9) EMR (Elastic Map Reduce): Equivalent to Hadoop on EC2 for big data processing
- 10) AWS Import/Export: Allow physical data transfer via shipping disks for high bandwidth data migration.

⑨ what are the key design objective of data center.

-
- 1) Scalability: Enable the infrastructure to scale up or down based on demand
 - 2) High availability and Reliability: Ensure continuous service availability with minimal downtime
 - 3) Security: Protect data, applications and infrastructure from threats.
 - 4) Energy efficiency: Reduce power consumption and carbon foot print
 - 5) Performance: Provides fast and reliable computing resources
 - 6) Cost efficiency: Minimize capital expenses, pay-as-you-go basis.
 - 7) Fault Tolerant: Ensure data and services are recoverable after a failure
 - 8) Manageability and Automation: Simplify operations and maintenance through automation

① Describe security of database services.

-
- 1) Database-as-a-Service is a key component of cloud computing, offering users scalable, managed database solutions.
 - 2) Entities involved in DBaaS security:
 - 1) Data owners: Individual or organization that generate and own the data
 - 2) Database users: Client accessing the DBaaS for queries and data retrieval
 - 3) Cloud Service providers: The companies that host and manage cloud databases
 - 4) Third Party Auditors: Independent entities that verify compliance & security
 - 3) Major security threats:
 - 1) Data Integrity and confidentiality risks:
 - a) Unauthorized access: Weak authentication expose database to unauthorized access
 - b) Insider threats: Superusers with excessive privileges can manipulate, leak or delete sensitive data.
 - 2) External attacks:
 - a) SQL injection: Attackers manipulate database queries to gain unauthorized access or modify data
 - b) Man in Middle attacks: Interception of data transmitted between users and DBaaS
 - 3) Multi-Tenancy Risks:
 - a) Data Leakage: Different users sharing same infrastructure increases the risk of data leaks
 - b) Illegal data Recovery: Improperly sanitized storage devices can allow retrieval of previously deleted data.

4) Availability Risk:

Resource exhaustion: Inefficient allocation of database resource can lead to system crashes

4) Security Measures for DBaaS

- 1) Strong authentication and access control
- 2) Data encryption and Secure storage
- 3) Secure Backup and Disaster Recovery
- 4) Auditing and Monitoring.

② Explain Security Risk posed by shared Images and Management OS

→ 1) shared Images

- a) Hidden viruses: Malware might be secretly embedded in the image
- b) Leftover data: Previous user's sensitive data may not be wiped
- c) Misconfiguration: Weak settings (open ports, weak password) can be reused
- d) Unauthorized access: If not restricted, others might use or copy your image
- e) No updates: Outdated software in images can be vulnerable to attacks.

2) Management OS:

- a) VM escape: Attackers can break out of their VM to attack others
- b) Central weakness: If the main OS is hacked, all VMs are at risk
- c) Insider threats: Admins with access can misuse it
- d) Vulnerabilities: Bugs in OS can be exploited by attackers
- e) Lack of isolation: Poor setup may allow one user's action to affect others.

③ Explain how virtual machine are secured.

③

→
1) Hypervisor Security: The hypervisor is the core software layer that manages VMs.

Types of Hypervisor:

Type 1 (Bare metal): Runs directly on hardware

Type 2 (Hosted): Runs on a Host OS

2) Virtual machine isolation: Isolation ensures that one compromised VM cannot affect others or the host system. It includes Resource isolation, Network segmentation.

3) Network Security: VMs can communicate over virtual network, which must be secured like physical ones, It includes firewalls, Intrusion detection, Virtual private network

4) VM Image Security: VM images can be a vector for attack if compromised

5) Guest OS security: Each VM runs as a guest OS that must be secured like any physical machine.

6) VM Lifecycle Management: Proper handling throughout the lifecycle of a VM prevents data leakage and misuse

7) Logging and Monitoring: Continuous monitoring helps detect and respond to threat quickly.

8) Backup and disaster Recovery: Regular backups ensure quickly recovery in case of an attack or failure, which includes encrypted backup, secure storage and Test restores.

④ Explain reputation system design options.

→

- 1) Centralized Reputation System: A centralized system relies on a single authority or server to collect, store and process reputation data for all users.
- 2) Decentralized Reputation System: In this design, reputation data is stored and processed across multiple nodes, with no central authority. Each participant maintains their own reputation scores.
- 3) Hybrid Reputation System: A hybrid system combines both centralized and decentralized models. Typically, reputation data is stored centrally, but peer-to-peer evaluations are used to influence final score.
- 4) Reputation Based on Feedback Mechanism: This system relies on user feedback or ratings after interacting with a service or user.
- 5) Reputation Based on Historical Behaviour: This system tracks the past behaviour of users or services. The reputation score is dynamically updated based on ongoing performance metrics.
- 6) Trust models in Reputation System: Trust models use algorithms or mathematical models to assign trustworthiness scores.
- 7) Reputation Based on Third party Evaluation: In this approach, a trusted third party organization evaluates the reputation of services or users in cloud.

⑤ What is XOAR? Discuss its role in providing trusted hypervisor support in cloud security.

→ XOAR is a micro-hypervisor framework. It focuses on security, trust, and flexibility. Designed to reduce the trusted Computing Base, minimizing the amount of code that must be trusted.

Role in cloud Security

- 1) Hypervisor Hardening: XOAR's small TCB means fewer bugs and vulnerabilities making cloud host more secure
- 2) Isolation: Ensures strong separation between VMs and between host and guest environments
- 3) Policy Enforcement: Fine-grained access control mechanism to prevent privilege escalation
- 4) Attestation and Trust: Provides evidence that the hypervisor is in a secure state important for sensitive workloads.
- 5) Resilience to attacks: The minimalist design is less exposed to attack vectors like VM escape, rootkits etc.

6) Describe the process and significance of a Privacy Impact Assessment in cloud computing.

→ Privacy Impact Assessment is a tool used to evaluate privacy risks in information systems

- 1) Planning and Preparation: Define the scope of the assessment, identify stakeholders, gather preliminary details about data flow and system architecture
- 2) Data Flow Analysis: Map out how personal data is collected, stored, processed, shared and disposed of.
- 3) Privacy Risk Identification: Analyze the type of personal data involved, determine risk such as, unauthorized access, data breaches, Profiling or surveillance.

- 4) Legal and Regulatory Compliance Check: Evaluate whether the system aligns with data protection act, general data protection Regulation, Industry-specific regulations.
- 5) Risk Evaluation and Mitigation: Assign risk levels to each identified issue. Data encryption, Access controls, Data minimization, Consent Mechanisms.
- 6) Documentation and Reporting: Prepare a detailed PIA report summarizing Project details, Privacy risks and impacts, Measures taken to reduce those risks.

Significance

- 1) Protects Individual Privacy
 - 2) Legal Compliance
 - 3) Build Trusts
 - 4) Risk Management
 - 5) Improves design.
- 7) Discuss various encryption techniques used for securing data in the cloud.

→

- 1) Classical Encryption: AES, RSA, ECC: standard encryption algorithms used to protect data at rest and in transit.
- 2) Homomorphic ~~Ency~~ Encryption: Allows ~~to~~ computation directly on encrypted data without decrypting it.
- 3) Fully Homomorphic Encryption: Enables arbitrary computations on encrypted data.
- 4) Searchable Encryption: Used for querying encrypted cloud databases without revealing the data.

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- a) Order-Preserving Encryption (OPE): Maintains the order of encrypted numeric values. Supports range queries on encrypted data.
 - b) Searchable Symmetric Encryption (SSE): Enables keyword searches on encrypted data. Protects both data and queries.
 - 5) Key Management and Encryption services:
 - AWS KMS: Manages and protects cryptographic keys.
 - Encryption SDK: Provided by cloud vendors for developers to integrate encryption into application.
 - 6) Threats Addressed by Encryption:
 - External Attacks: Hackers accessing CSP infrastructure.
 - Internal Threats: Malicious insiders.
 - Network attacks: Man-in-the-middle attacks during data transmission.

Module 5

Cloud Computing

① What are the various system issues for running a typical parallel program in either parallel or distributed manner?

→

1) Communication Overhead:

Parallel system: may have lower communication latency due to shared memory

Distributed system: must send data over a network, leading to higher latency and bandwidth constraints.

2) Synchronization and Coordination: Ensuring that multiple processes or threads coordinate properly is critical

3) Data Distribution and locality: How data is divided among processes affects performance.

4) Load Balancing: uneven workload distribution causes some nodes/threads to be idle while others are overloaded.

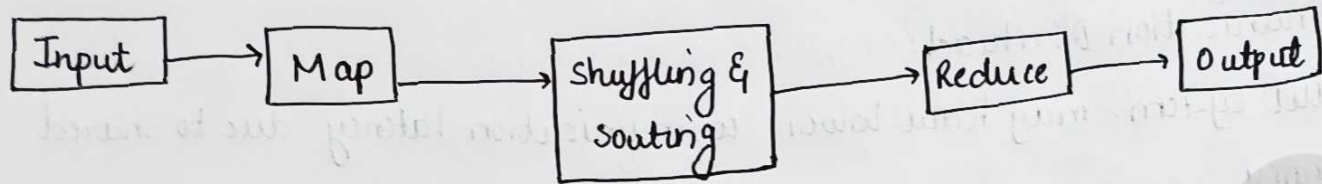
5) Fault Tolerant and Reliability: In distributed systems, nodes or network link can fail, system must handle failures gracefully.

6) Scalability: The ability of the system to maintain performance as more resources are added.

7) Resource Management: Effective use of CPU, memory, network usage, and storage.

8) Security Issues: Data transmission over network introduces concerns about data integrity, confidentiality and authentication.

② ~~next~~ with a neat diagram explain the data flow in running a MapReduce job at various task trackers using Hadoop library.



The data processed by MapReduce should be stored in HDFS, which divides the data into blocks and store distributedly

Step1: One block is processed by one mapper at a time. In the mapper, a developer can specify his own business logic as per the requirements. In this manner, Map runs on all the nodes of the cluster and process data blocks in parallel.

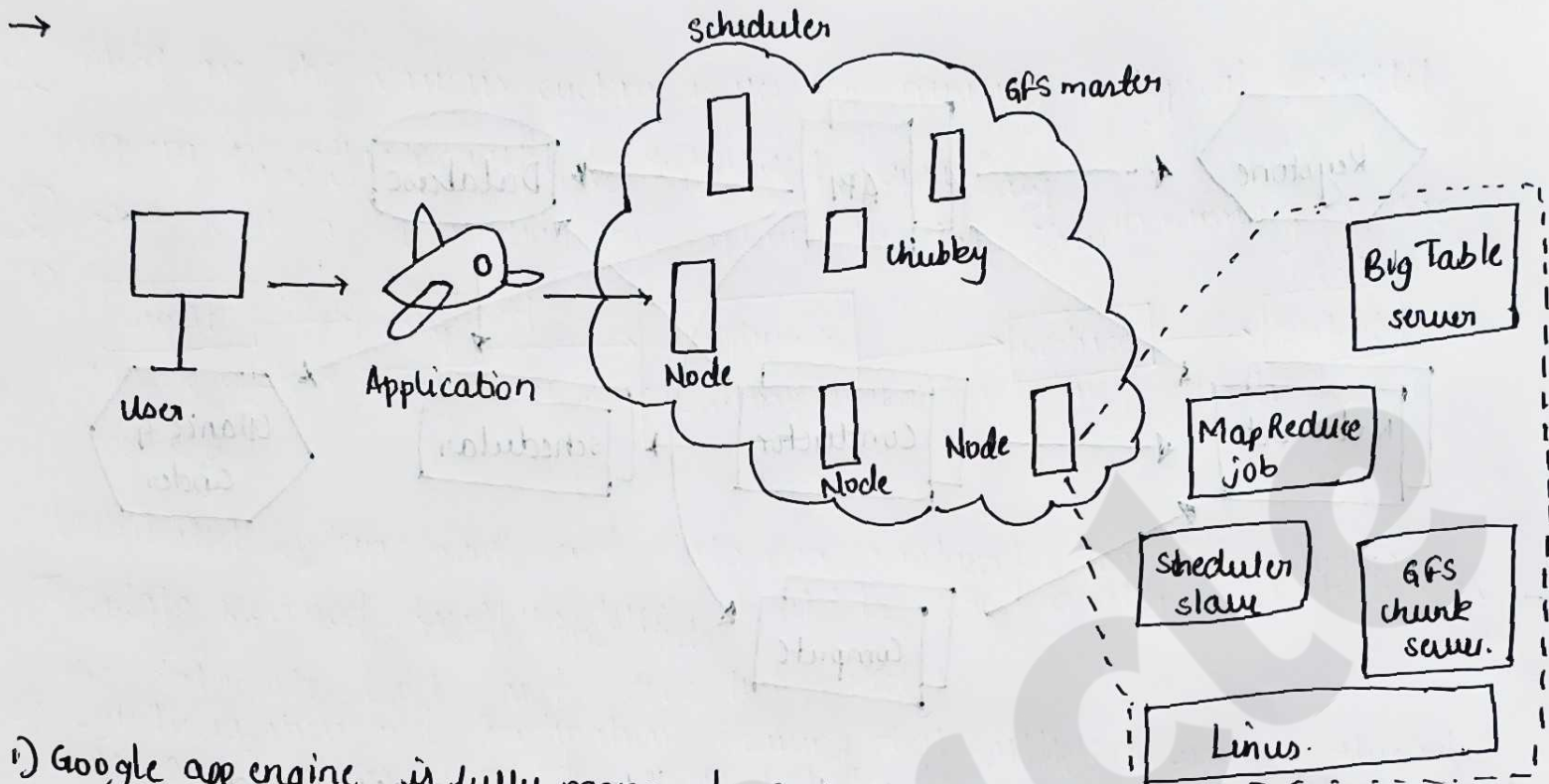
Step2: Output of Mapper also known as intermediate output is written to the local disk. An output of mapper is not stored in HDFS as this is temporary data and writing on HDFS will create unnecessary many copies.

Step3: Output of mapper is shuffled to reducer node. The shuffling / copying is a physical movement of data which is done over the network.

Step4: Once all the mappers are finished and their output is shuffled on reducer nodes then this intermediate output is merged and sorted. which is then provided as input to reduce phase.

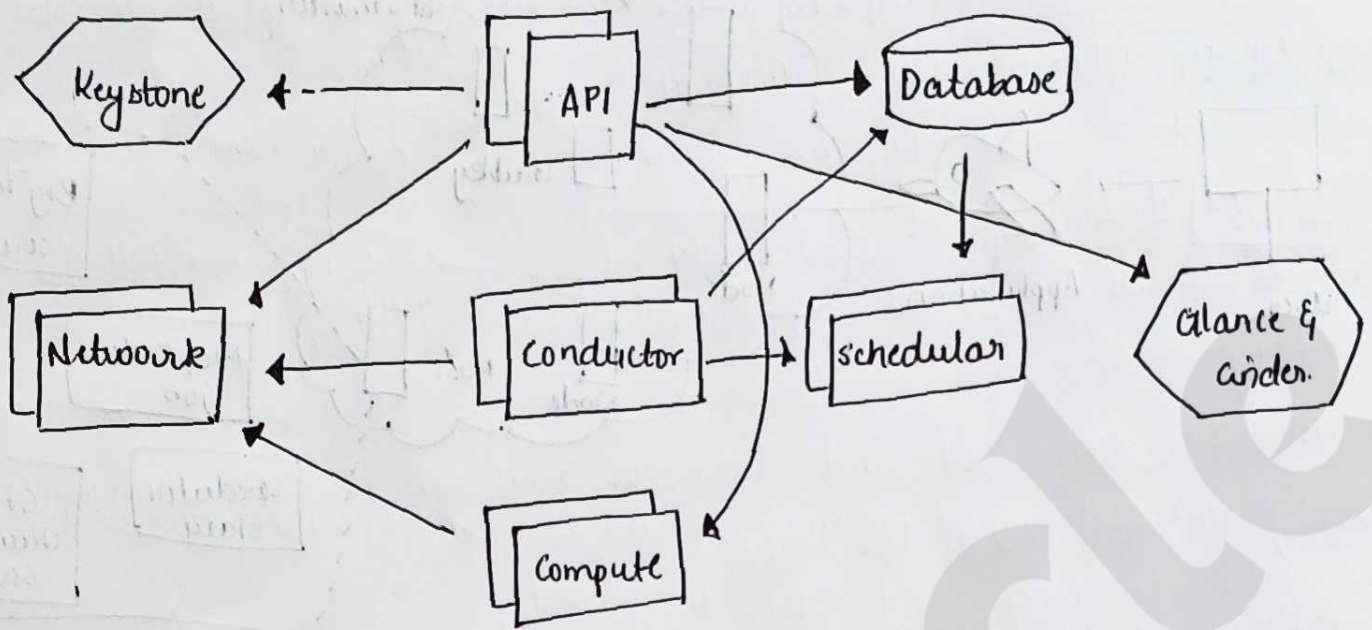
Step5: Reduce is the second phase of processing where the user can specify his own custom business logic as per the requirements. An input to a reducer is provided from all the mappers. An output of reducer is the final output which is written on HDFS

③ Discuss Programming the Google App Engine



- 1) Google app engine is fully managed Platform as a service used for building and hosting scalable web applications on google infrastructure.
- 2) It dynamically scales web application as traffic demand changes, ensuring efficient resource usage and high availability.
- 3) GAE support multiple programming language like python, Java, Go and PHP.
- 4) The app engine SDK allows developers to emulate the production environment on local machine and later deploy their apps easily with cost control quotes.
- 5) GAE provides numerous in-built services including cron job, queues, scalable datastores, communication tools.
- 6) It offers a secure and high-performance execution environment with general features.
- 7) Third party services and helper libraries are integrated via partnership enabling app to perform extended task beyond core functionality.

④ with neat diagram explain Openstack Nova system architecture



- 1) NOVA is an openstack component responsible for managing and provisioning virtual Machine instances, similar to AWS EC2, but for private clouds.
- 2) It supports multiple hypervisor and virtualization technologies such as KVM, Hyperv, VMware.
- 3) Nova interacts with other open stack services like
 - a) Keystone for authentication
 - b) Glance for image services
 - c) Neutron for network provisioning
 - d) Cinder for providing volumes to VM instance.
- 4) Nova is developed using python, and uses libraries like
 - a) Gevent for networking
 - b) sqlalchemy for database interactions.
- 5) It follows a horizontally scalable architecture, meaning the workload is distributed across multiple servers instead of relying on a single machine.

6) Nova uses SQL databases to store information, which are shared logically by its components.

7) Core components in Nova architecture like

1) DB: Central SQL database

2) API: Handles HTTP request and interacts with other components

3) Scheduler: Allocates instances to hosts.

4) Compute: Manages VMs and hypervisors.

5) Conductor: Coordinates complex tasks and acts as a DB proxy.

⑤ What are the emerging cloud software environments?

→ This section introduces open source and research oriented cloud platform and tools designed to support cloud programming, VM management, storage and data processing

1) Eucalyptus: ~~original~~ Allows users to upload/manage VM images bundle root file systems and register them for deployment. Emulates Amazon EC2 and S3. Available in both open source and commercial version

2) Nimbus: Enables leasing remote resources via VM deployment, open source IaaS cloud solution

Cumulus: S3-compatible storage system with quotas

3) Open Nebula: Manages full VM life cycle and dynamic networking. uses libvirt API, CLI, and cloud drivers for hybrid cloud, open source IaaS cloud solution.

4) Sector / Sphere:

Sector: wide-area DFS with replica placement based on network topology

Sphere: works with Sector to process data using user-defined function

5) Open stack: Open source cloud platform started by Rackspace and NASA, Message based, shared nothing architecture. Supports replication, failure isolation and heterogeneous storage.

6) Aneka cloud: Aneka is a cloud platform for developing and running parallel and distributed applications built on .NET but supports Linux via Mono.

⑥ Explain the different parallel and distributed programming models used in cloud computing.

→

1) MapReduce Model: A programming paradigm for processing large dataset in parallel.

Map: Processes input data into key-value pair.

Reduce: Aggregates intermediate results based on keys.

2) Twister: Extension of MapReduce for iterative tasks. Keeps data in memory between iterations.

3) MPI (Message passing Interface): Low-level model for tightly coupled parallel applications.

4) Dryad and Dryad LINQ:

Dryad: A general purpose engine from Microsoft using DAG to represent computation.

DryadLINQ: High level interface that allow C# developers to write distributed programs using LINQ.

5) Sawzall: A scripting language developed by Google, analyzing very large scale log data.

6) Pig Latin: High level data flow language developed by Yahoo, best for structured data processing.

⑦ Explain the steps involved in deploying a web application on App engine with automatic scaling.

→

1) Develop the web application: Build your app locally using

Java: Eclipse with GAE plugin, GWT for rich interfaces

Python: Lightweight webapp framework.

2) Configure the app for deployment: Create the app.yaml file to define configuration such as

1) Application ID

2) Runtime environment

3) Automatic scaling settings

4) URL handlers

3) Define the data storage: Use google app engine datastore for ~~storing~~ storing structured NoSQL data: Entities with key-value properties, Memcache for performance, Blobstore for large files.

4) Local Testing and Debugging: use GAE's local development server, to test your application. Debug and verify data interactions, routing and functionality.

5) Deploy to App Engine: use Google cloud SDK or IDE tools to ~~display~~ deploy the app

6) Automatic scaling and Hosting: Once deployed, app engine automatically scales your application based on traffic

7) Manage and Monitor: use the google cloud Console to

1) View logs

2) Track performance

3) check resource usage and billing.

8) Explain the key features of cloud and grid computing platforms and compare them.

→

cloud computing platforms:

- 1) Elasticity: Automatically scale resources up/down based on demand
- 2) Virtualization: Use VMs to isolate workloads for different user/application
- 3) On demand self service: Users can provision resources without human interaction
- 4) Massive data storage: Distributed file system like HDFS, S3.
- 5) Massive data Processing: Supports parallel processing.

Grid Computing platforms

- 1) Resource sharing: Integrates resources from multiple domains
- 2) Decentralized Management: Resources are managed independently across different organization.
- 3) Batch oriented Processing: Best suited for large scale scientific computations submitted as jobs
- 4) Data Transport: Uses standard protocols and supports large file transfers
- 5) Security Frameworks: Uses certificate and role-based access.

* cloud Computing

Grid Computing

- | | |
|---|--|
| 1) Typically managed by cloud providers | Resources owned and shared by multiple organizations |
| 2) Automatic and dynamic resource scaling | Static or manual resource scaling |
| 3) User friendly | More technical |
| 4) Built in security | Custom Certificates. |
| 5) Programming Models: SaaS, IaaS, PaaS. | MPI, GridRPC, SAGA |